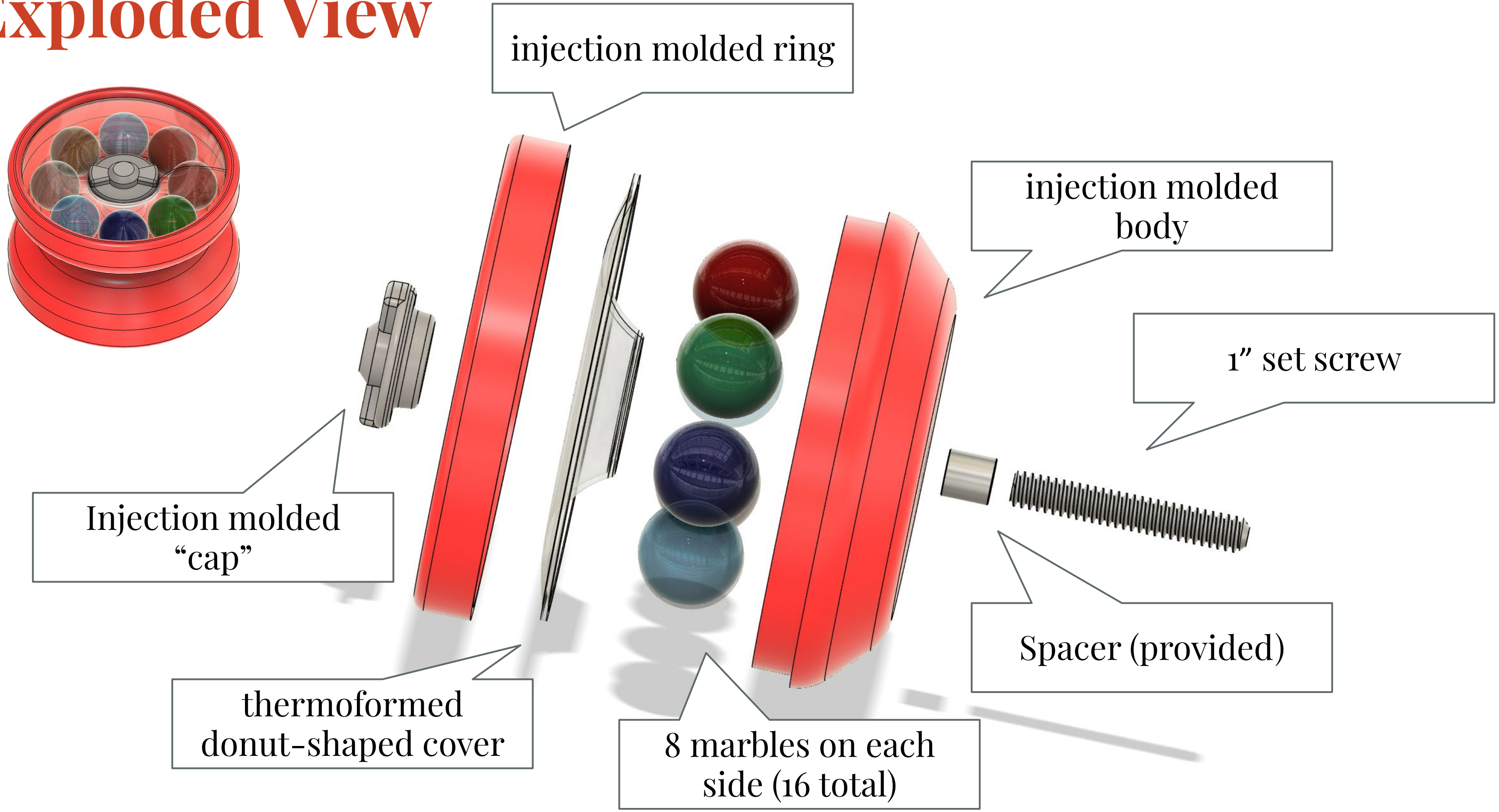
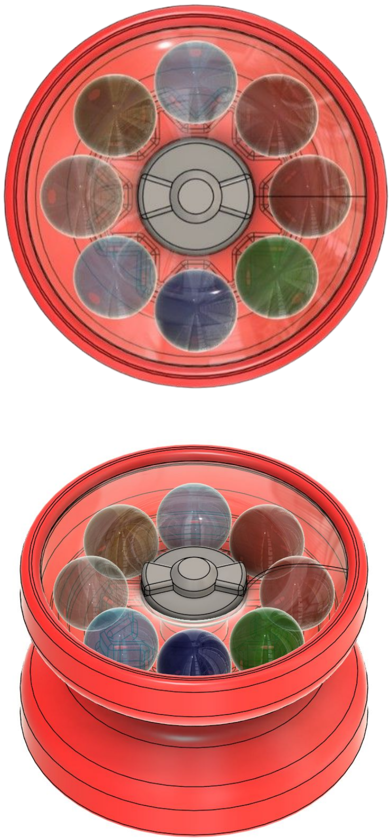
Gumball Yo-Yo Design Review

Team K: The Gumbalumballers

Exploded View



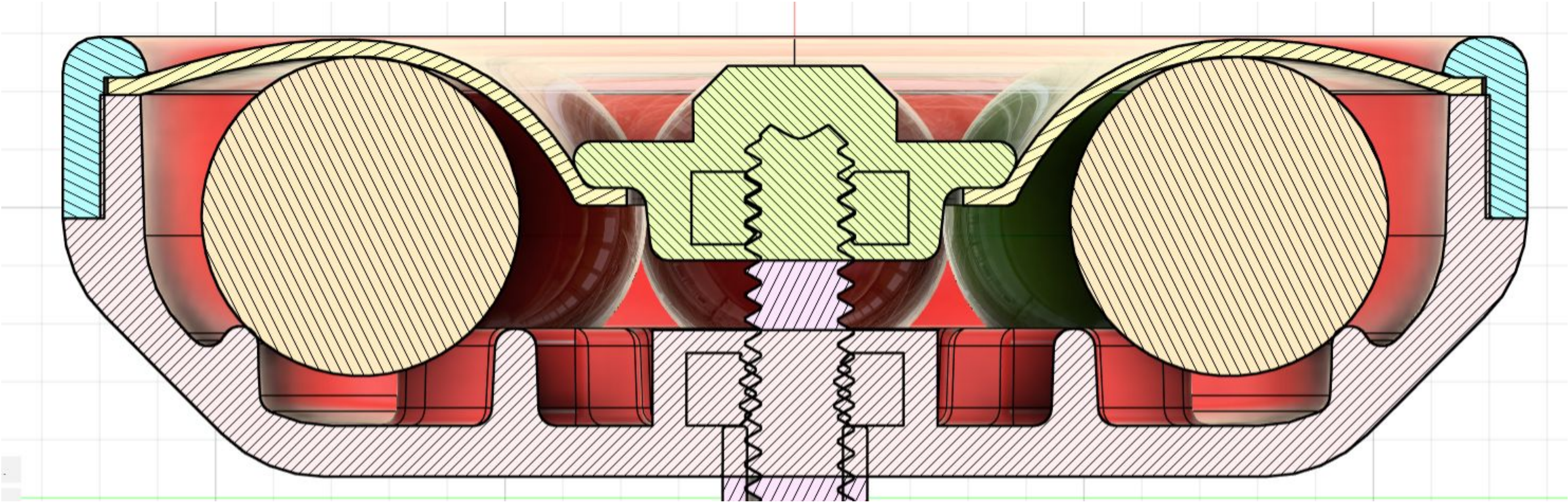
Yo-yo Parts



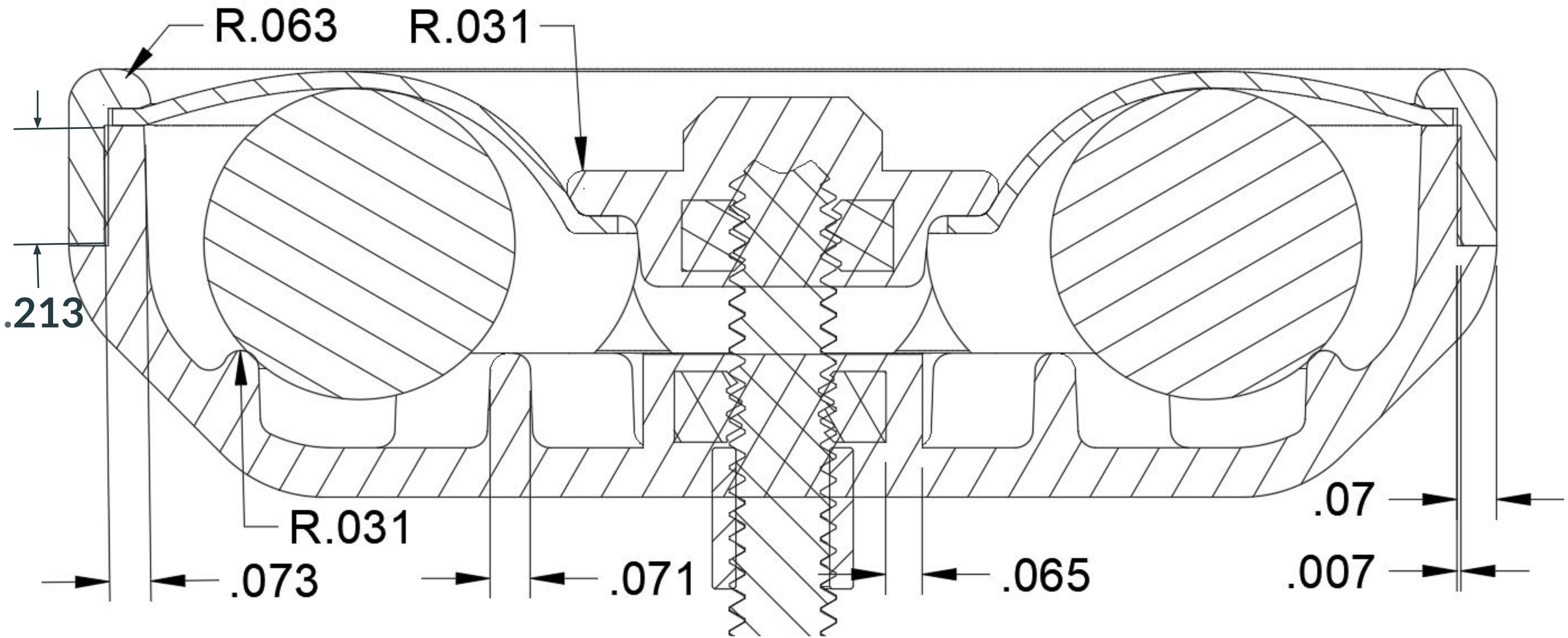
Part Name	# Per Yo-Yo	# Needed	# on hand	Manufacturing Method or Source
Body	2	100	0	IM (overmolded nut)
Ring	2	100	0	IM
Cap	2	100	0	IM (overmolded nut)
Cover (“donut”)	2	100	2	Thermoformed
Marbles	16	800	1000	Purchased (1000x)
Set Screw	1	50	50	Purchased
Hex nut	4	200	300	Provided
Spacer	1	50	50	Provided

Total IM parts: 300 (200 with overmolded nuts)
Total IM molds: 6 (3 core/cavity combos)

Cross Section & Press Fit



Cross Section & Press Fit



Press Fit Engineering Analysis

- **Target:** 0.014 in
- **Successful Range:** 0.013-0.016 in (estimated)
- **Strategic Approach**: Started with reference recommendation from Canvas, then measured yo-yo's with press fits that felt very tight using calipers; the average interference was **~0.014 in**.
    - Started with slightly smaller than recommended 0.015" because it's easier to increase the interference later by removing material from the molds
- **Link to shrinkage modeling:**
    - CAD is pre-scaled (**body 1.0215**, **ring 1.0173**)

Shrinkage Analysis

Scaling Parameters to accound for Shrinkage

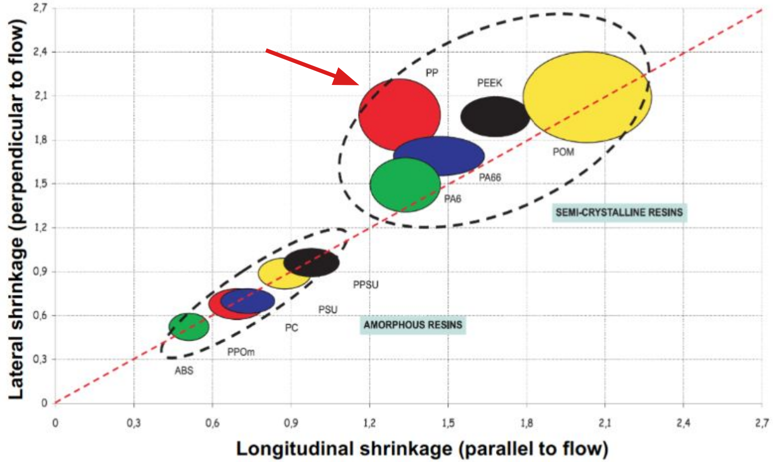
shrinkage from simulation (MoldFlow) for body:	2.4%		
shrinkage amount for body:	2.1%	scaling factor for body:	1.0215 (use in Fusion360)
shrinkage from simulation (MoldFlow) for ring:	1.3%		
shrinkage amount for ring:	1.7%	scaling factor for ring:	1.0173 (use in Fusion360)

Updated diameter values due to shrinkage modeling:

SCALED Body, Nominal Diameter:	2.554	[in]
SCAELD Ring, Nominal Diameter:	2.513	[in]

How we chose shrinkage parameters:

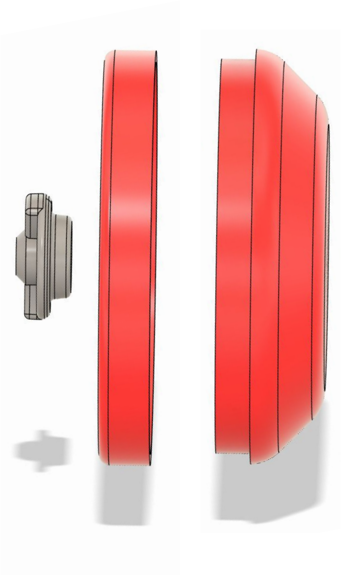
- Conducted IM simulations in Fusion
- Looked at typical lateral shrinkage for PP (chart →)
- Selected values between the typical levels (1.8-2.1%) and the simulation results for the body and core

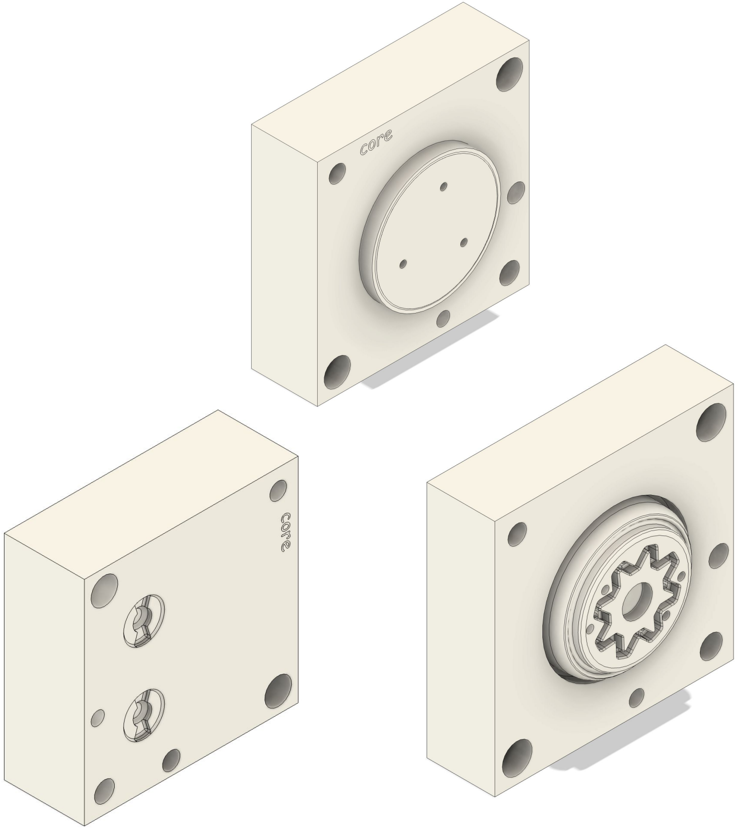


http://www.lati.com/pdf/technical_data/dimensional-molding-shrinkages.pdf

Molds & CAM

3 IM Parts





- ❑ Run time: 6.5 hrs
  - 50% Body
  - 25% Ring
  - 25% Cap
- ❑ Machinability
- ❑ Include all necessary features
- ❑ Peer & Instructor Reviewed

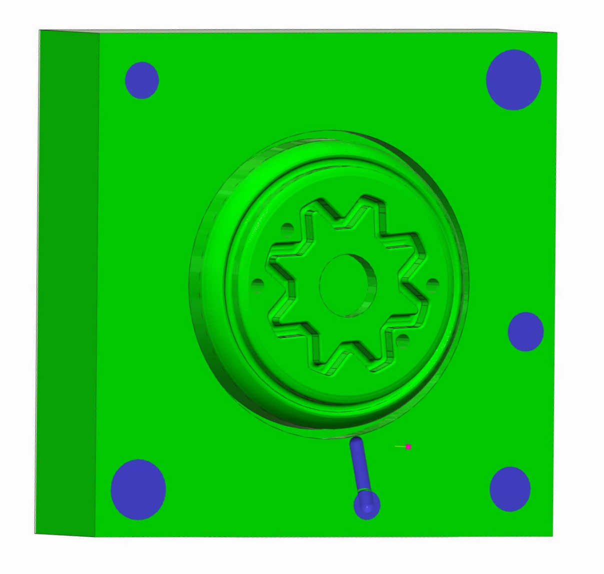
Yo-Yo Body CAM

Body Core CAM

Run Time: 02:47:36

16 Operations

9 unique tools



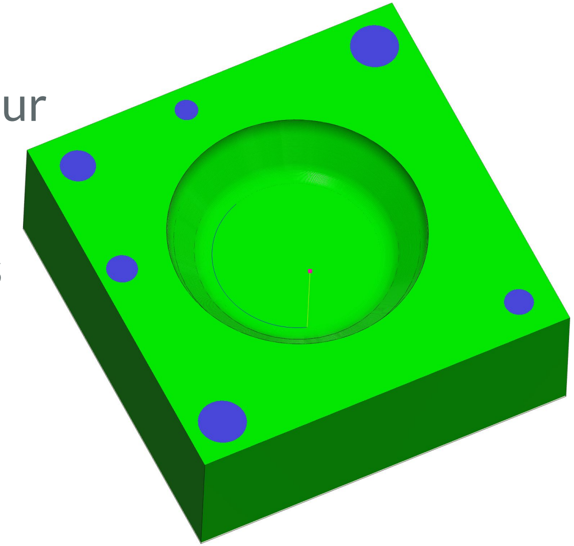
Body Cavity CAM

Run Time: 01:36:32

5 Operations

- Face
- 3D Adaptive Clearing
- 3D Flat
- 3D Contour
- Scallop

4 unique tools



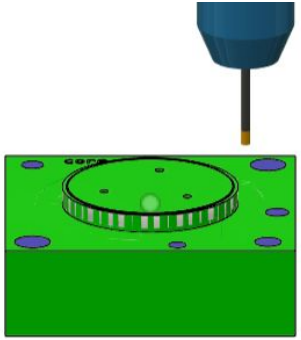
Yo-Yo Ring CAM Summary

Ring Core CAM

Run Time: 13:00

7 Operations

6 Distinct Tools



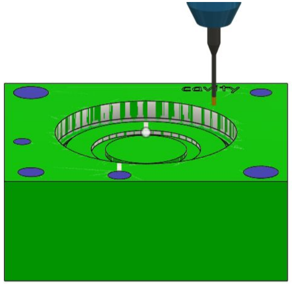
- Face
- Center Drill
- Deep Drill
- Flat
- Bore (finishing pass for press fit)
- 2D Contour
- Spiral

Body Cavity CAM

Run Time: 29:35

5 Operations

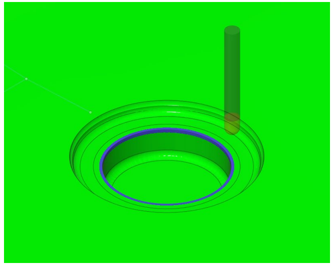
5 Distinct Tools



- Face
- Adaptive Clear
- Bore
- 3D Contour
- 2D Contour
- 2D Contour

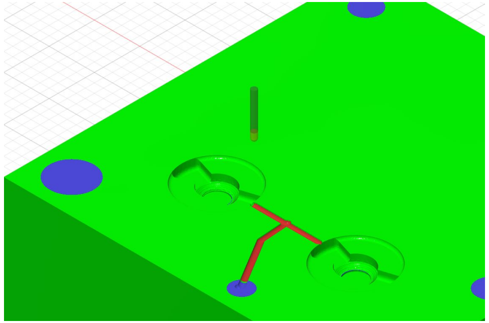
Yo-Yo Cap CAM Summary

Cavity Mold CAM



- **4 tools**
  - Facing
  - ¼ ” flat
  - 1/16” taper
  - 1/16” ball
- **8 operations**
- **Runtime:** 16 minutes

Core Mold CAM



- **4 tools**
  - Facing
  - ⅛” flat
  - 1/16” flat
  - 1/16” ball
- **12 operations**
- **Runtime:** 51 minutes

To do: Fix runner, drill ejector pin holes

High Risk Features & Plan

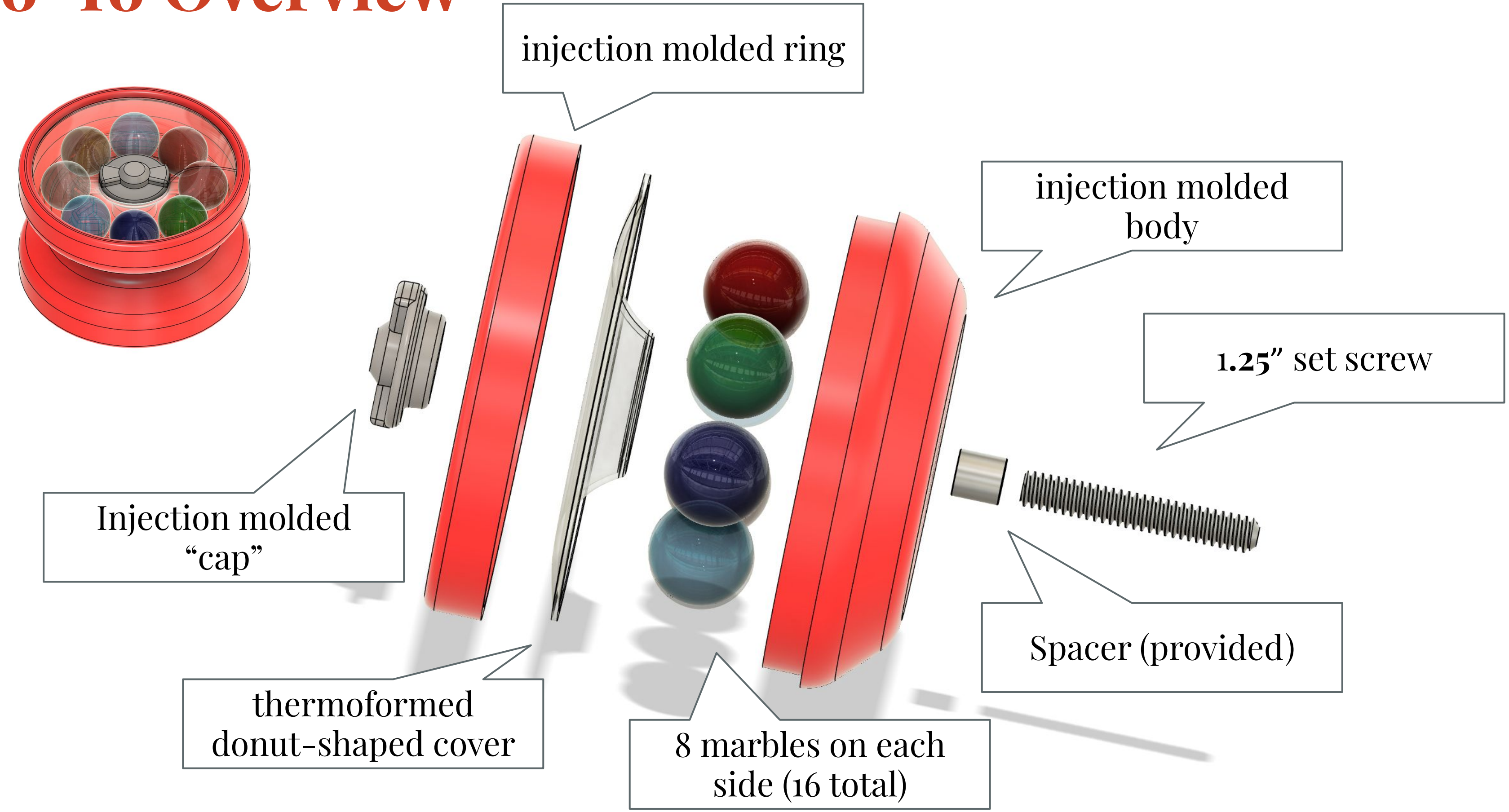
- **High risk features:**
  - Marble containment
  - Extra overmolded IM part (cap)
  - Hole in thermoformed part
  - Long set screw system
- **Fallback plan:**
  - Omit marbles and/or cap
  - Do not cut hole in thermoformed part
  - Switch to standard set screw center
  - → Viable yo-yo with body + ring + thermoformed cover
    - Can redesign thermoformed part if desired

Thank You!

Gumball Yo-Yo Manufacturing Review

Team K: The Gumbalumballers

Yo-Yo Overview



Finished Yo-Yo

Drop Test

Without cap

With cap

Progress Update

- All molds are final
  - Ring molds: Machined 2x
  - Body molds: Machined 3x
  - Cap molds: Machined core 2x, cavity once
- IM parameters are tuned for all parts
- Shoulder bolt overmolding implemented for cap & body
- ~75 thermoformed parts made

TODO:

- Make final rings, bodies, caps (mix colors)
- Make the remaining thermoformed parts
- Assemble

Press Fit Engineering Analysis

Design Target

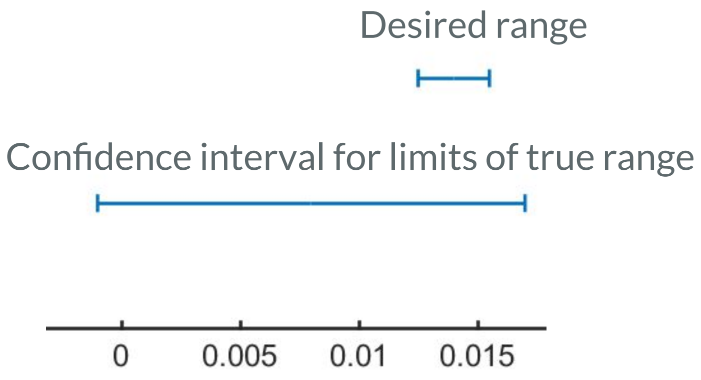
- **Diametric Interference:** 0.014 in
  - Based off mean diametric interference measurements from yoyo's in lab with strong interference fits
- **Successful Range:** 0.013-0.016 in (estimated)

Results

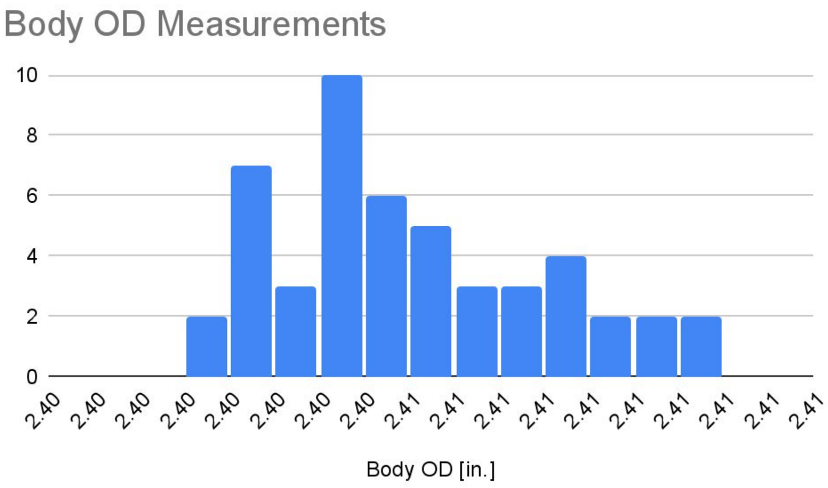
- **Mean Diametric Interference:** 0.008 in.
- **Standard Deviation:** 0.009 in.
- **95% Confidence Interval for random interference:**

0.016 to - 0.014 in.

  - Measurement is inconclusive



Press Fit Engineering Analysis

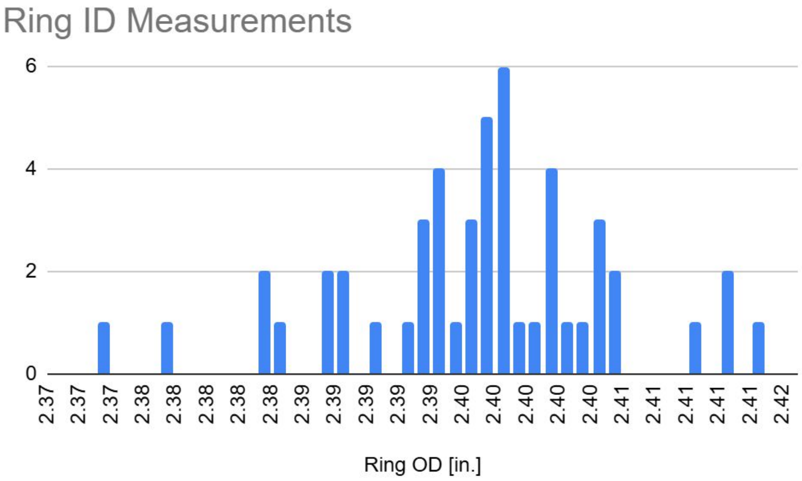


Body OD Distribution

μ = 2.405 in.

σ = 0.003 in.

N = 50



Ring ID Distribution

μ = 2.397 in.

σ = 0.008 in.

N = 50

Press Fit Engineering Analysis

Mismatched Targets & Results

- Imperfect calculation of body & ring shrinkage
- Thin parts deform when you measure them, and are hard to measure accurately and consistently

CMM (Actual) Measurements for Mold

- **Body:** 2.455 in.
- **Ring:** 2.432 in.

Target (Design) Measurements for Mold

- **Body:** 2.456 in.
- **Ring:** 2.431 in.

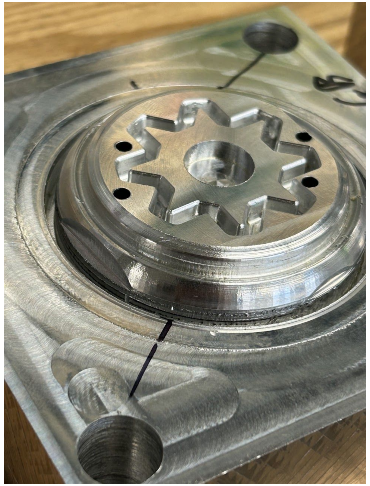
Press Fit - Lessons Learned

- Press fit worked very well in actuality
  - Passed initial drop test
- Starting with a smaller designed press fit was good, gives room for adjustment if necessary
- Shrinkage calculation is difficult to accurately determine
- Cap & long set screw system was necessary to hold weight of marbles during drop test

Still thinking of ways to accurately & precisely measure rings...

Manufacturing

- Ring molds: Machined 2x
- Body molds: Machined 3x
- Cap molds: Machined core 2x, cavity once
- Total Manufacturing time: **20 hrs**
- Mold blanks used: 16



DFM - Body Mold

Mold Manufacturing Challenges

- Machined 3x
  - 1st time: end mill chatter created burs; caused deburring mistake
  - 2nd time: unwanted fillet from tapered end mill on OD face of body led to poor interference fit
- Core took ~4 hours to set up and machine

CAM, Mold, and Design Solutions

- Used 2” facing end mill to clear out most of the stock and reorder ops
  - Decreased core and cavity machine time by 30 min
- Increased wall thickness of CAD to give taper end mill more clearance
- Added finishing facing op to remove burs

DFM - Ring Mold

Mold Manufacturing Challenges

- Machined 2x
    - Runner + ejector pin holes machined on wrong side

CAM, Mold, and Design Solutions

- Ring CAM required a special end mill with longer flute length (0.375” vs 0.0890”)
- Ring molds do not have tapered faces due to press fit
- Ejects from an internal runner rather than the part because we don’t have large enough flat faces that line up with ejector pins

DFM - Cap Mold

Mold Manufacturing Challenges

- Machined core 2x, cavity once
  - Incorrect reaming and machining runner error

CAM, Mold, and Design Solutions

- Ejector pins moved to center for aesthetic purposes

DFM - Injection Molding

I.M. Parameter Optimization

- Started with parameters in the IM booklet and adjusted based off resulting IM part
  - I.e. reduced injection pressure to be just high enough that molds fill completely, to prevent flashing

Body			Ring		
machine (Boy/Arburg):	Boy	[mm]	machine (Boy/Arburg):	Boy	[mm]
dosage stroke:	35.5	[mm]	dosage stroke:	20	[mm]
switch over stroke:	7.5	[mm]	switch over stroke:	7.5	[mm]
injection pressure:	900	[psi]	injection pressure:	800	[psi]
packing pressure:	850	[psi]	packing pressure:	500	[psi]
packing time:	15	[s]	packing time:	15	[s]
cooling time:	30	[s]	cooling time:	10	[s]
ejector pin length:	5.375	[in]	ejector pin length:	5.625	[in]
sprue length:	5	[in]	sprue length:	5	[in]
shim thickness:	0.01	[in]	shim thickness:	0	[in]
dye color:	1.5% - #28 0.5% - #3		dye color:	1.5% - 28 0.5% - 3	[#]
dye mass:	7.5g - #28 2.5g - #3	[g]	dye mass:	7.5g - #28 2.5g - #3	[g]
PP batch mass:	500	[g]	PP batch mass:	500	[g]

DFM - I.M. Parameter Optimization

DFM - Thermoforming

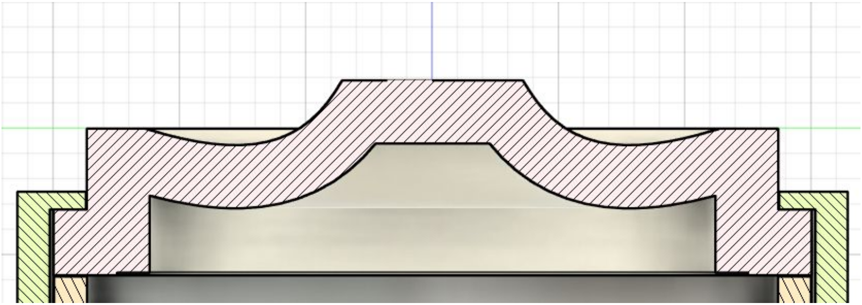
Thermoforming Parameter Tuning

- Started with default parameters
- Tuned for part clarity
  - Decreased heating time to lower sensitivity to fine mold features
- Tuned for low cycle time
  - Used fan to increase cooling time

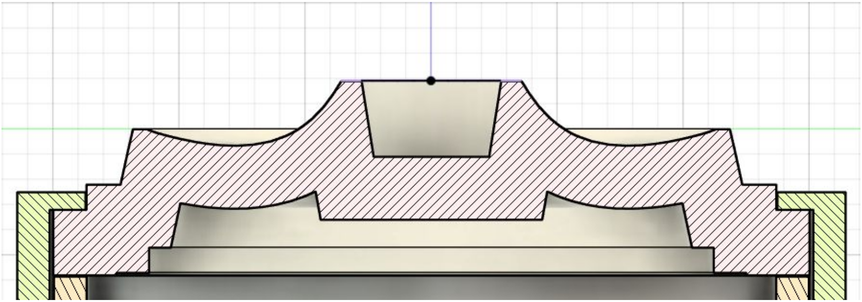
heating time:	5.5	[s]
forming time:	10	[s]
platen delay:	2	[s]
platen return:	1.5	[s]
eject:	1	[s]
vacuum delay:	0	[s]
fan delay:	4	[s]

DFM - Thermoforming Mold

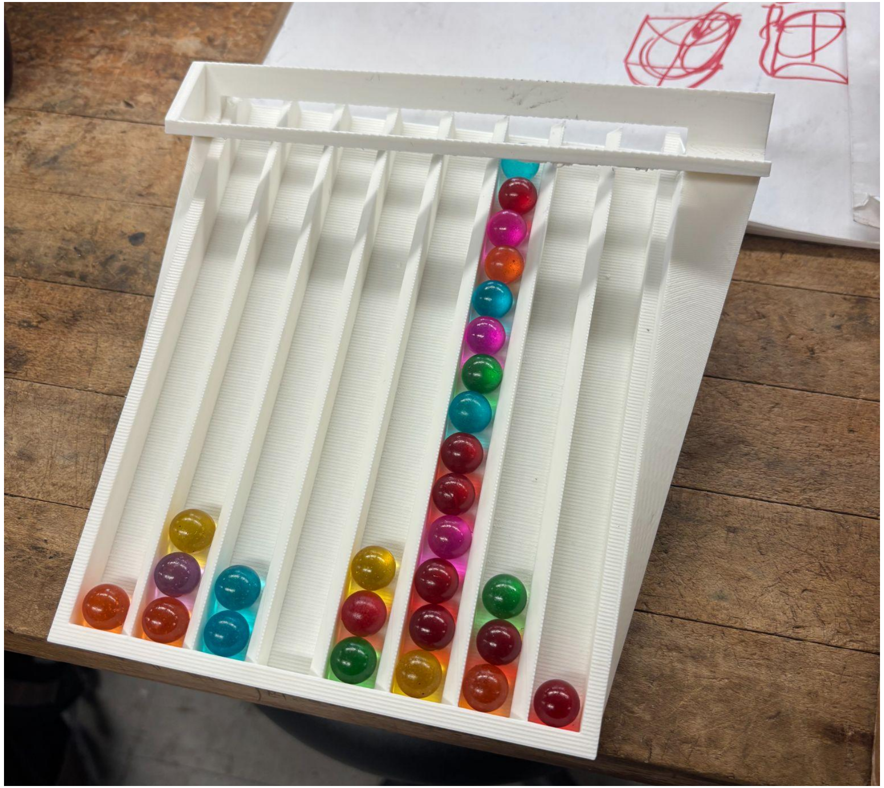
- Initial design:
  - Hard to align arbor press and hole punch to center holes
  - Hole location important for marble containment



- Revised design:
  - Tapers added so that punch fits over center taper and arbor press fits within outer taper



DFM - The Marble Factory



- Marble factory has loose tolerances
- Easily sort 1000 marbles into ~6 thou groups

Remaining Challenges

- Ring doesn’t eject properly from mold
  - Solution 1 : Easy to pull out by hand
  - Solution 2: File down the gates to the ejection ring more
- Marble vibration decreases yo-yo ability (dampens rotation)
  - Solution 1: Shallower thermoformed part for tighter retention
  - Solution 2: Sort out smaller diameter marbles

Planning & Organization

- Did not use backup plan, still pursuing original design
  - We were able to finish machining all of the molds
  - Each group member shared responsibilities for machining & lab objectives
- Interesting strategic decisions
      - Used special ⅛" end mill with longer flute length for ring mold (instead of altering CAD)
      - Used 2" face mill to clear most of stock for body core
      - Used another machine shop (Area 51) when all other machines were booked
      - Special tool to remove shoulder bolts from caps
      - Sorting tool to separate marbles by size

Thank You!

Appendix - Tools